Advanced SQL Null Functions

Question One

1)
```
SELECT name,
       employee_id,
IF NULL (salary, 0) AS salary_with_default
FROM Employees;
```

Employees

employee_id	name	salary_with_default
1	Alice	5000
2	Bob	0
3	Charlie	7000
4	Dana	0

2
```
SELECT employee_id,
       name,
IF NULL (department, Not Assigned) AS department_name
FROM Employees
```

employee_id	name	department_name
1	Alice	HR
2	Bob	IT
3	Charlie	Not Assigned
4	Dana	Finance

3)
```
SELECT order_id,
       customer_id
FROM Orders
WHERE customer_id IS NULL;
```

Orders

order_id	customer_id
103	NULL

Question 4

```
SELECT    Order_id ,
          customer_id ,
          ~~delivery status~~
          COALESCE (delivery_date, 'Pending') AS delivery_status
FROM  orders ;
```

Orders

Order_id	customer_id	delivery status
101	201	2024-12-01
102	202	Pending
103	NULL	2024-12-03

Question 5

```
SELECT    student_id ,
          name ,
          COALESCE (grade , 0) AS final_grade
FROM  students ;
```

Students

Student_id	name	final_grade
1	Ethan	85
2	Maya	0
3	Olivia	90

Question 6

```
SELECT   COUNT (student_id ) AS not_graded_count
FROM   Students
WHERE   grade = 'NULL' ;
```

Students

not_graded_count
1

Question 7

```
SELECT   product_id ,
~~name~~   name
~~COALESCE (price - discount, 0) AS final_price~~
~~FROM products~~ (Price - COALESCE (discount ;0) AS finalprice
From   Products ;
```

Product_id	name	final_price
501	keyboard	25
502	Mouse	10
503	Monitor	100

Question 8

```
SELECT   ~~customer_id~~ name ,
COUNT (*) AS missing_email_count
FROM   customers
WHERE   email IS NULL
```

customers

missing_email_count
2

```
SELECT customer_id,
       name
       COALESCE(email, 'No email') AS email_display
FROM customers
```

customers

customer_id	name	email_display
1	Linda	No email
2	Joseph	joseph@mail.com
3	Nia	No email

Question 10

```
SELECT payment_id,
       status,
       COALESCE(method, 'Unknown') AS method_display_status
FROM Payments;
```

payments

Payment_id	status	method_display_status
301	Credit	Unknown
302	PayPal	Sucess
303	Unknown	Failed

Question 11

```
SELECT item_id,
       item_name
       COALESCE(quantity, 0) AS quantity_checked
FROM Inventory;
```

Inventory

Item_id	item_name	quantity_checked
1	Pen	0
2	Notebook	150
3	Eraser	0

Question 12

```
SELECT emp-id ,
COALESCE (bonus , commission) AS first_reward
FROM Employees-Extra
LIMIT 1 ;
```

Employees_Extra

emp_id	first_reward
1	300

Question 13

~~SELECT~~

~~COUNT (COALESCE (room),~~ rom

Question 13

```
SELECT
        COUNT (room) AS no_room_count
FROM Classes
WHERE room = 'NULL'
```

Classes

no_room_count
2

Question 14

```
SELECT student_id ,
       ~~status~~ date
COALESCE (status , Not Marked) AS attendance_status
FROM Attendance ;
```

Attendance

student_id	date	attendance_status
1	2025-04-01	Not Marked
2	2025-04-01	Present
3	2025-04-01	Absent

Question 15

```
SELECT account_id
    COALESCE (account_type, Unknown) AS type_display
    COALESCE (balance ; 0) AS balance checked
FROM Bank_Accounts ;
```

Bank_Accounts

account_id	type_display	balance checked
A1	Saving	0
A2	Current	5000
A3	Unknown	2000

Question 16

```
SELECT project_id ,
       title
COALESCE (start_date, TBD) AS start_display
FROM Projects ;
```

Projects

project_id	title	start_display
1	Website Revamp	2025-01-10
2	Mobile App	TBD
3	Data Migration	TBD

Question 17

```
SELECT    review_id ,
          product_id
  COALESCE (comment , No comment) AS comment_display,
  COALESCE ( rating , 0 ) AS ~~display_0~~ rating_display
FROM      Reviews
```

Reviews

review_id	product_id	comment_display	rating_display
1	501	Great product	4
2	502	No comment	0
3	503	Works fine	3

Question 18

```
SELECT    suppler_id ,
          name
  COALESCE (phone , alt_phone , 'No Contact') as contact_number
FROM      Suppliers
```

Suppliers

supplier_id	name	contact_number
1	Global Goods	123456879
2	Best Supplies	78765431
3	Value Source	No contact

Question 19

```
SELECT user-id,
 COALESCE (theme, 'Light') AS theme-set,
 COALESCE (language, 'English') AS language-set,
 COALESCE (timezone, 'UTC') AS timezone-set
FROM user_settings;
```

User settings

user-id	theme-set	language-set	timezone-set
1	Light	English	UTC
2	Dark	English	UTC +1
3	Light	English	UTC

Question 20

```
SELECT record-id,
       machine-id,
COALESCE (issue, 'Unkown Issue') AS issue-log,
COALESCE (technician, 'Not Assigned') AS technician name
FROM Maintenance ;
```

Mantenance

record-id	machine-id	issue-log	technician-name
1	M101	Overheating	Not Assigned
2	M102	Unkown Issue	Not Assigned
3	M103	Jammed	Alex